

WASP-50b

R-1208

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_191010 · created 2026-07-31T07:32:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458766.887 +/- 0.02 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.049
Transit depth (Rp/Rs)^2	2.47 +/- 1.53 [%]
Orbital Inclination (inc)	89.59 +/- 2.9
Airmass coefficient 1 (a1)	0.92 +/- 0.18
Airmass coefficient 2 (a2)	0.05 +/- 0.13
Scatter in the residuals of the lightcurve fit is	19.68 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	7.5
Transit Duration (day)	0.076 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.049 vs 0.1404 (deviation 0.28 σ)
Duration fit vs published	0.076 d vs 0.0754 d (deviation 0.03 σ)
Mid-time O-C	-47.7 min
Depth SNR	2.7
Residual scatter	19.68 %

Light curve

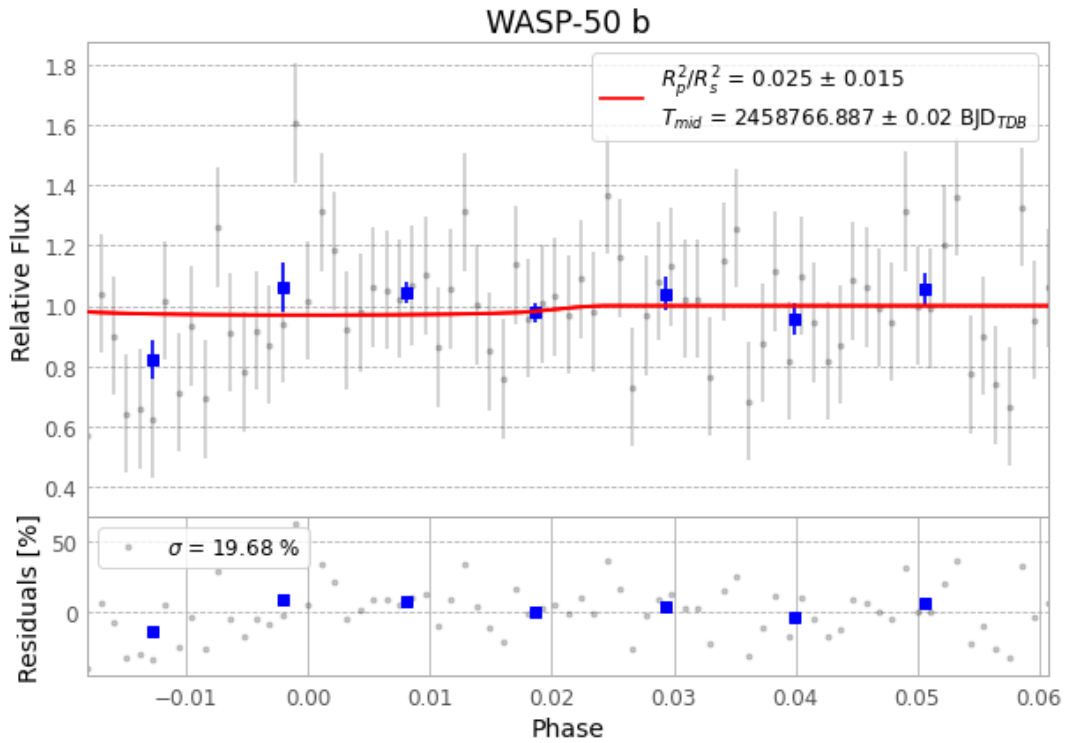


Figure 1 — FinalLightCurve_WASP-50 b_2019-10-10.png (SHA-256 fdb068cfc2b928f9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [199, 354], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6c2657bb312488ae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-50191010081812.FITS → WASP-50191010120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-191010.log (d786b517c835...), FinalLightCurve_WASP-50 b_2019-10-10.png (fdb068cfc2b9...), FinalLightCurve_WASP-50 b_2019-10-10.csv (450bec4b6152...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:32Z.